Independent Cross-Validation of *Forced versus unforced Navier–Stokes*

独立交叉验证报告

Prepared by an AI research assistant, with two independent referees (AI) and machine checks

2 November 2026 · audited version: `c63c0a4` · revised version: `c873aa2`

Abstract

English. We report an independent audit of the 20-page paper *Forced versus unforced Navier–Stokes: what a prescribed force can and cannot prove* (commit `c63c0a4`). The audit had three independent components: (i) two referee agents — fresh model instances, prompted as hostile referees with different mandates, and given only the files — reviewed every numbered statement and proof; (ii) machine checks re-derived the algebra symbolically (SymPy) and reran the numerical verification in a clean directory; (iii) the literature invoked by the paper was checked against primary or survey sources.

The audit produced **15 findings**: 2 serious errors (a missing hypothesis that made the two-dimensional "complete proof" circular, and a maximal-regularity lemma stated in a form that is false at the endpoint), 4 further errors, 1 factual error about the literature, 3 imprecisions, 1 missing term, 1 missing hypothesis, 1 claim supported only by floating-point arithmetic, 1 overreach and 1 overstatement of novelty. All 15 have been corrected in commit `c873aa2`. After revision the paper's *conclusions* survive, but three claims had to be weakened: the prescription theorem is elementary (folklore), the price law is an identity for exactly self-similar solutions rather than a general lower bound, and one corollary (discrete self-similarity) was false and has been deleted. We also record what the audit *confirmed* independently, and what it could not check at all.

中文摘要. 本报告是对 20 页论文《受迫与无受迫纳维–斯托克斯：给定的外力能证明什么、不能证明什么》（提交 `c63c0a4`）的独立交叉验证。验证由三部分组成：(i) 两个独立的"审稿人"模型实例（只给文件、不给作者的推理过程，分别以"严苛审稿人"和"专挑具体步骤"的角色提示）；(ii) 机器核验（SymPy 符号重推 + 在干净目录重跑数值脚本）；(iii) 文献核对（Clay 官方四种提法、2026 年三篇 arXiv 预印本、NRS 与 ESŠ 两个经典定理）。

共发现 **15 处问题**：2 处严重错误（二维"完整证明"缺假设导致循环；极大正则性引理写成了端点处不成立的形式）、4 处其他错误、1 处文献事实错误、3 处不精确、1 处漏项、1 处缺假设、1 处仅靠浮点运算支撑的断言、1 处越界推论、1 处新颖性夸大。全部 15 处已在 `c873aa2` 中修正。修正后论文的主要结论仍然成立，但有三处必须弱化：指定力定理是初等的（属 folklore）、代价定律只对"精确自相似"成立且是等式、以及一个推论（离散自相似）本身是错的、已删除。报告同时列出审计确认的部分与完全无法核验的部分。

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1 Scope, versions, artefacts

Item	Version audited (v1)	Revised (v2)
Paper	`c63c0a4` (20 pp.)	`c873aa2` (20 pp.)
Numerical script	`paper/verify.py`	unchanged
Chinese lecture notes	`fdb26c2`	minor edits (four Clay formulations)
Referees	2 fresh model instances	—

The paper under audit claims complete proofs of: 2D global regularity (Thm 5.1); Prodi–Serrin (Thm 4.2) and Beale–Kato–Majda (Thm 4.6); a "prescription theorem" for the forced problem (Thm 7.1); a manufactured blow-up (Cor 7.3); and a "price law" for exactly self-similar blow-up (Thm 8.1), with two corollaries. It *quotes* (does not prove) the 2026 theorems of Constantin–Ignatova–Vicol and Ingimarson–Kukavica–Xu, and the autoformalisation result of Bastounis–Circelli–Hansen.

2 Method

2.1 Two independent referees

Two referee agents were instantiated as separate model instances with no access to the author's conversation, notes or reasoning; each received only the LaTeX source and the verification scripts. Their mandates differed deliberately:

- **R1 (journal referee).** "Hostile but fair referee for a top PDE journal. Verify mathematics step by step; you may not say 'standard' without checking. Give a verdict per numbered item, then the five most serious problems." R1 ran 29 review iterations and also fetched the three arXiv abstracts and the Clay statement.
- **R2 (targeted auditor).** "Be terse and concrete; compact output." R2 was given a specific list of the seven places where a referee would expect trouble: the sign in Kato's inequality, the trace-space issue in the maximal-regularity lemma, the bootstrap to a local sup norm, the pressure sign in the prescription theorem, the H^s scaling exponents, whether the pressure in Theorem 8.1 is necessarily self-similar, and the $L^{2/3}$ claim in Remark 8.4.

Neither referee saw the other's report; the two reports were written independently and agree on the two serious errors.

2.2 Machine checks

- **Symbolic (SymPy).** Six identities re-derived from their definitions: the chain rule $\frac{d}{dt}[\lambda U(\lambda x)] = \dot\lambda(U + y\cdot\nabla U)$ for three profiles; $\dot\lambda/\lambda^3 = a$; the exponent identity $\lambda^{3-3/q} = (2a(T-t))^{-\frac{3}{2}(1-\frac{1}{q})}$; Kato's identity $\Delta|w|^p - p|w|^{p-2}w\Delta w = p(p-1)|w|^{p-2}|\nabla w|^2$ for $p = 2, 4$; the Prodi–Serrin/Young exponent identities; the vorticity of the Gaussian profile; and $\mathrm{curl}((u\cdot\nabla)u) = (u\cdot\nabla)\omega$ in 2D. All residuals are exactly zero (Appendix B).
- **Clean-room numerics.** `paper/verify.py` was copied to an empty directory and re-run; the output reproduced Appendix B to all digits ($\max|\operatorname{div} f| = 1.511\times10^{-10}$, scaling error 4.491×10^{-12}, $(T-t)^{3/2}\|f\|_\infty = 1.153384$ at all three times).
- **Figures.** The 2D simulation behind the lecture notes was re-run from `notes/make_figs.py`; the reported invariants ($\max|\omega|$ decreasing from 2.152 to 1.950, $\|\omega\|_2$ ratio 0.9217) are consistent with $\|\omega\|_p$ being non-increasing.

2.3 Source checks

Checked against primary or survey sources, not against the paper's summary: the Clay problem has *four* official formulations, two without and two with a smooth external force; the three 2026 arXiv abstracts (2609.20803, 2610.25531, 2610.08144) and their exact hypotheses; the exclusion of non-trivial backward self-similar solutions in $W^{1,2}$ by Nečas–Růžička–Šverák (via Tsai's survey, arXiv:1802.00038, §2); and the $L^\infty_t L^3$ regularity theorem of Escauriaza–Seregin–Šverák.

3 Findings at a glance

Table 1 lists all 15 findings. "v1 verdict" refers to the audited version; the last column records what was done in v2.

Table 1: All findings of the audit.

ID	Item	v1	Issue	Correction in v2
F1	Prop 2.1	**ERROR**	$x \mapsto \lambda x$ does not preserve the torus unless $\lambda \in \mathbb{N}$, so u_λ is not defined on $\mathbb{T}^3$ for general λ	require $\lambda \in \mathbb{N}$ on $\mathbb{T}^3$
F2	Thm 3.1 (proof)	**ERROR**	no eigenbasis of the Stokes operator on $\mathbb{R}^3$ (continuous spectrum); and $\|B(u,u)\|_{H^{-2}} \le C\|u\|\|\nabla u\|$ has the wrong exponents	prove on $\mathbb{T}^3$/bounded domain, cit Heywood's basis for $\mathbb{R}^3$; correc bound to $\|u\|^{1/2}\|\nabla u\|^{3/2}$
F3	Rem 4.3	**ERROR** (factual)	claimed the $L^\infty_t L^3$ criterion is "open for large data"; it is a theorem (ESŠ 2003)	replaced by the correct statement; th open range is $q < 3$
F4	Rem 4.5	**ERROR** (minor)	the L^q identity omits the forcing term	added $\int f \cdot u\|u\|^{q-2}$ with its bound
F5	Thm 5.1	**ERROR** (serious)	$u_0 \in H^1$ does not give $\omega_0 \in L^p$ for $p > 2$, so Step 2 is void; and $\sup_{[0,T]} \|\nabla u\|_p < \infty$ is false at $t = 0$	theorem restated for $u_0 \in H^2$; boun over $[\delta, T]$; H^1 by approximatio (new Remark 5.5)
F6	Lemma 5.2	**IMPRECISION**	hypothesis $\omega_0 \in L^p$ missing	added
F7	Lemma 5.4	**ERROR** (serious)	a pointwise-in-t bound with $F \in L^\infty_t L^p$ is not available from maximal regularity (only L^q_t, $q < \infty$); $\int_0^t \|Ae^{\nu\tau A}\|\, d\tau$ diverges	lemma restated in L^q form; Step rewritten with $q = p$, giving ∇u $L^p_t L^\infty \subset L^1_t$
F8	Thm 6.1	**IMPRECISION**	slice statement gave only $\mathcal{H}^2$	strengthened to $\mathcal{H}^1$
F9	Thm 7.1	**IMPRECISION**	pressure normalization missing, so the converse is false (p is determined only up to $c(t)$, and the reconstructed force is $f - \nabla c$)	normalization added; converse quali fied
F10	Cor 7.2	**ERROR**	"bijection onto all smooth forces" is false: the image lies in $\{\operatorname{div} f = 0\}$ (counterexample $f = \nabla\phi$, $u_0 = 0$); injectivity was proved with a 2D inequality	restricted to divergence-free forces 3D-valid energy proof of injectivity
F11	Cor 7.3	unproved	$F \not\equiv 0$ (which makes $\|f\|_\infty \to \infty$) was asserted from floating-point arithmetic	proved: $F \equiv 0$ would give a non trivial backward self-similar solutio in $W^{1,2}$, excluded by NRS 1996
F12	eq. (8.5)	**ERROR**	exponent written as $-(3 - 3/q)$ instead of $-\frac{3}{2}(1 - \frac{1}{q})$	corrected
F13	Cor 8.3	**ERROR** (false)	discrete self-similarity: self-similarity at a sequence does not determine $\partial_t u$, and a depends on k; not a consequence of Thm 8.1	corollary deleted
F14	Cor 9.2	over-reach	concluded that f is not C^2-bounded, which does not follow; IKX hypothesis dropped	analyticity excluded unconditionally vanishing only in the C^2-bounde case; IKX hypothesis restored; Pro 10.1(b) softened
F15	Thm 7.1 (novelty)	overstatement	the prescription identity is elementary and folklore (the standard "any path can be driven" observation)	novelty claim withdrawn; kept as precise statement plus the control in terpretation

4 The two serious errors, in detail

4.1 F5: the 2D "complete proof" was circular

In v1 the section opens with "$u_0 \in H^1_\sigma(\mathbb{T}^2)$", and Lemma 5.2 gives

$$\|\omega(t)\|_p \le \|\omega_0\|_p + \int_0^t \|g(s)\|_p \, ds. \tag{1}$$

For $p > 2$ the right-hand side involves $\|\omega_0\|_p$, but $u_0 \in H^1$ only gives $\omega_0 \in L^2$. Hence $M_p = \infty$ for every $p > 2$, Step 3 ($B(u,u) \in L^\infty_t L^p$) never starts, and nothing beyond $p = 2$ is proved. The displayed conclusion $\sup_{t\in[0,T]} \|\nabla u(t)\|_p < \infty$ is also false at $t = 0$ for $p > 2$, since $\nabla u_0 \notin L^p$ in general.

Fix. The theorem is now stated for $u_0 \in H^2_\sigma$, which gives $\omega_0 \in H^1 \subset L^p$ for all $p < \infty$, and the bound is asserted on $[\delta, T]$ (and on $[0,T]$ if $u_0 \in W^{1,p}$). The H^1 case is recovered by approximation, using the difference estimate of Proposition 5.6, whose constant depends only on $\|\nabla v\|_{L^2_t L^2}$ and is therefore uniform by the energy inequality (new Remark 5.5). This approximation does not use (1) at $p > 2$.

Why it matters. The failure is not cosmetic: the entire bootstrap is driven by $\omega \in L^\infty_t L^p$ with $p > 2$, precisely so that $W^{1,p} \subset L^\infty$ in two dimensions. Without the hypothesis the proof has no content.

4.2 F7: maximal regularity cannot be quoted at $q = \infty$

v1 asserted, for $F \in L^\infty(0,T;L^p)$,

$$\sup_{\delta\le t\le T} \big(\|u(t)\|_{W^{2,p}} + \|\partial_t u(t)\|_p\big) \le C(\delta,p,\nu,T)\big(\|u_0\|_p + \|F\|_{L^\infty(0,T;L^p)}\big).$$

Giga–Sohr give $\|u\|_{L^q_t W^{2,p}} + \|\partial_t u\|_{L^q_t L^p} \le C_q \|F\|_{L^q_t L^p}$ for $1 < q < \infty$ only. The endpoint $q = \infty$ fails because $\int_0^t \|A e^{\nu\tau A}\|_{p\to p}\, d\tau = \infty$: no pointwise-in-time bound uniform in F can hold.

Fix. The lemma is restated in L^q form, and Step 4 takes $q = p$: since $B(u,u) + f \in L^\infty_t L^p \subset L^p_t L^p$, maximal regularity gives $u \in L^p(0,T^*;W^{2,p})$, hence $\nabla u \in L^p(0,T^*;L^\infty)$ and, since $T^* < \infty$,

$$\int_0^{T^*} \|\nabla u(t)\|_\infty \, dt \le (T^*)^{1-1/p} \|\nabla u\|_{L^p_t L^\infty} < \infty,$$

which is exactly what Beale–Kato–Majda needs. The revised proof is shorter: the δ-truncation is no longer needed.

4.3 F10, F12, F13: the remaining structural errors

Cor 7.2. The v1 claim "the map (7.2) is a bijection … onto the smooth forces for which (1.1) has a smooth solution" is false, because the construction always produces $\operatorname{div} f = 0$: with $f = \nabla\phi$ and $u_0 = 0$, $(u,p) = (0,\phi)$ is a smooth solution but $\operatorname{div} f = \Delta\phi \neq 0$. The revised statement restricts to divergence-free forces (harmless: a gradient is absorbed into the pressure) and replaces the 2D inequality used in the injectivity proof by the 3D estimate $\|w\|_4 \le C\|w\|^{1/4}\|\nabla w\|^{3/4}$.

Equation (8.5). With $\lambda = (2a(T-t))^{-1/2}$, $\lambda^{3-3/q} = (2a(T-t))^{-\frac{3}{2}(1-\frac{1}{q})}$. The v1 exponent $-(3-3/q)$ is off by a factor $3/2$ in the power of $(T-t)$; the corrected identity passes check C2 of Appendix B.

Cor 8.3 (deleted). v1 asserted that if $u(\cdot,t_k) = \lambda_k U(\lambda_k x)$ along a sequence $t_k \uparrow T$, then $\limsup_k (T-t_k)^{3/2}\|f(t_k)\|_\infty = (2a)^{-3/2}\|F\|_\infty$. This does not follow: the theorem computes f from $\partial_t u$, which self-similarity at isolated times does not determine, and a is not defined by the data (instead $a_k = 1/(2\lambda_k^2(T-t_k))$, which may depend on k). What survives is the correct statement already contained in Theorem 8.1: if the solution is self-similar *on an interval* with a fixed profile, then $\dot\lambda/\lambda^3 \equiv a$ and the price law holds identically in t.

5 What the audit confirmed independently

The referees agreed, after independent recomputation, on the following; these are the parts of the paper that did *not* change.

- **Prodi–Serrin (Thm 4.2).** The Hölder exponents $(q,\ 2q/(q-2),\ 2)$, the Gagliardo–Nirenberg exponent $\theta = 3/q$, Young's inequality with $\sigma = 2q/(q+3)$, $\sigma' = 2q/(q-3)$, and the equivalence $2q/(q-3) \le p \iff 2/p + 3/q \le 1$ were checked line by line.
- **Beale–Kato–Majda (Thm 4.6).** The commutation estimate, the Gronwall step and the continuation argument are correct as stated.
- **Kato's inequality.** The direction is correct ($s \mapsto |s|^p$ is convex), and $\int \Delta\omega\, \omega|\omega|^{p-2} \le \frac{1}{p}\int \Delta|\omega|^p = 0$ is legitimate; SymPy check C3 verifies the identity for $p = 2, 4$.
- **The 2D bootstrap** ($\omega \in L^\infty_t L^p \Rightarrow \nabla u \in L^\infty_t L^p \Rightarrow u \in L^\infty \Rightarrow$ bootstrap $\Rightarrow$ BKM) is correct once the hypothesis on ω_0 is in place.
- **The prescription identity.** The pressure sign convention, the divergence computation $\operatorname{div} f = \operatorname{div}((v\cdot\nabla)v) + \Delta q = 0$, and the formula for F were confirmed; the clean-room run reproduces $\max|\operatorname{div} f| = 1.5 \times 10^{-10}$.
- **The price law.** The chain rule, $\dot\lambda = a\lambda^3$, the λ^3 scaling of F, and the resulting identity (with the corrected exponent) are correct; SymPy checks C1–C2 confirm the two identities on which everything rests.
- **The exponents in Cor 7.3.** $\|v\|_{\dot H^s} = \lambda^{s-1/2}\|U\|_{\dot H^s}$, $\|v\|_\infty = \lambda\|U\|_\infty$, and the threshold $s = 1/2$ are correct.

6 What the audit could not check

1. **The quoted 2026 theorems.** The statements of Constantin–Ignatova–Vicol and of Ingimarson–Kukavica–Xu were checked against their abstracts, but their proofs were not reproved. Corollary 9.2 is only as good as those theorems.

2. **The 2026 manuscript.** Neither the 166-page manuscript nor the Lean development was audited; the paper only *quotes* the reported facts (166 pages, 3 withdrawn manuscripts, 721/373, 42% formalised).

3. **The numerics as evidence.** The checks verify identities; they do not verify any *a priori* bound. T3's chain-rule test has a finite-difference error of 2.6×10^{-6} and therefore cannot detect an error of relative size 10^{-6}.

4. **The 2D simulation.** It is a 192^2 spectral run, not a resolution study; the monotone L^p decay is a theorem, and the simulation only illustrates it.

5. **Human review.** No human referee has read the revised paper.

7 Recommendations for auditing machine-written mathematics

The exercise suggests a cheap, reproducible protocol.

R1. Two referees with different mandates. One verifies statements globally; one is given the list of places where the argument is most likely to fail. Agreement is weak evidence; *disagreement* is a reliable pointer to a real problem.

R2. Check hypotheses against the space, not the name. F5 and F6 both came from "$u_0 \in H^1$" being silently upgraded to "$\omega_0 \in L^p$". Ask, for every lemma, which norm of the *data* appears on the right-hand side.

R3. Distrust endpoint exponents. F7 is the generic failure mode of quoting maximal regularity at $q = \infty$; any "$\sup_t$" estimate derived from an L^q_t theorem should be re-derived.

R4. Re-derive every exponent symbolically. F12 (an error of a factor 3/2) would survive any amount of prose review and is caught in one line by SymPy.

R5. Separate "numerically verified" from "proved". F11's claim was true, but the support was floating point; it became a theorem only after a literature argument was found.

R6. Re-run numerics in a clean directory. One minute; it rules out path dependence, stale output and caching.

R7. Check the source of every quoted theorem. F3 was a false statement about the literature that no amount of local checking would reveal.

8 中文版报告

本节是前面各节的中文表述，可独立阅读。

一、做了什么

对论文《受迫与无受迫纳维–斯托克斯：给定的外力能证明什么、不能证明什么》（20 页，提交 `c63c0a4`）做了一次三路交叉验证：

1. **两个独立“审稿人”模型实例**。只给 LaTeX 源文件和脚本，不给作者的推理过程。第一位以“顶级期刊的严苛审稿人”为角色，逐条核对定理与证明；第二位被明确告知七个最容易出错的位置（Kato 不等号方向、极大正则性的端点、局部上界 bootstrap、压力符号、H^s 指数、压力是否必然自相似、$L^{2/3}$ 断言），要求逐条判定。两人互不通气。

2. **机器核验**。SymPy 从定义出发重推六组恒等式（链式法则、$\dot\lambda = a\lambda^3$、指数换算、Kato 恒等式、Prodi–Serrin/Young 指数、二维涡量恒等式）；把 `verify.py` 复制到空目录重跑；重跑二维涡量模拟。

3. **文献核对**。Clay 官方四种提法（两种无外力、两种带光滑外力）；三篇 2026 年 arXiv 预印本的摘要与假设；NRS 关于“$W^{1,2}$ 中不存在非平凡后向自相似解”的结论（经 Tsai 综述核实）；ESŠ 的 $L^\infty_t L^3$ 正则性定理。

二、结论：15 处问题，全部已修

编号	位置	判定	问题与修正
F1	命题 2.1	错误	环面上 $x \mapsto \lambda x$ 只在 $\lambda \in \mathbb{N}$ 时保持周期；已加假设
F2	定理 3.1 证明	错误	$\mathbb{R}^3$ 上 Stokes 算子没有特征基；$B(u,u)$ 的 H^{-2} 界指数写错；已改
F3	注 4.3	事实错误	误称 $L^\infty_t L^3$ 判据“大初值未解决”；实际是 ESŠ 2003 的定理；已更正
F4	注 4.5	错误（小）	L^q 恒等式漏掉外力项；已补
F5	定理 5.1	严重错误	$u_0 \in H^1$ 推不出 $\omega_0 \in L^p$（$p>2$），第二步失效；且 $[0,T]$ 上的上界在 $t=0$ 不成立
F6	引理 5.2	不精确	缺 $\omega_0 \in L^p$ 假设；已补
F7	引理 5.4	严重错误	把极大正则性写成 L^∞_t 的逐点形式（端点不成立）；已改为 L^q 形式并重写第四步
F8	定理 6.1	不精确	切片只给到 $\mathcal{H}^2$；已加强为 $\mathcal{H}^1$
F9	定理 7.1	不精确	缺压力归一化，逆命题不成立（p 只确定到 $c(t)$，重构出的外力是 $f - \nabla c$）
F10	推论 7.2	错误	“到所有光滑外力的双射”不成立（像集只含无散外力，反例 $f = \nabla\phi$）；单射性用了二维不等式
F11	推论 7.3	未证明	$F \not\equiv 0$ 只由浮点运算支撑；现已用 NRS 定理证明
F12	式 (8.5)	错误	指数写成 $-(3-3/q)$，应为 $-\frac{3}{2}(1-\frac{1}{q})$
F13	推论 8.3	错误（假）	离散自相似的版本不成立（$\partial_t u$ 不被确定，a 依赖于 k）；已删除
F14	推论 9.2	越界	推出了“f 在 C^2 中无界”，推不出；已改为：解析性被无条件排除，恒零只在 C^2 有界时排除
F15	定理 7.1 新颖性	夸大	指定力恒等式是初等的、属 folklore；已撤回新颖性表述

三、两处严重错误是什么

F5（二维证明曾循环）：引理 5.2 给的是 $\|\omega(t)\|_p \le \|\omega_0\|_p + \int \|g\|_p$，右边需要 $\|\omega_0\|_p$；但 $u_0 \in H^1$ 只给 $\omega_0 \in L^2$。于是 $p>2$ 时 $M_p = \infty$，第三步（$B(u,u) \in L^\infty_t L^p$）根本启动不了。修正：定理改为对 $u_0 \in H^2$ 陈述（此时 $\omega_0 \in H^1 \subset L^p$），上界写在 $[\delta, T]$ 上；H^1 情形用差分解的稳定性（命题 5.6）做逼近，新增注 5.5。

F7（极大正则性用錯端点）：原引理声称对 $F \in L_t^\infty L^p$ 有逐点（$\sup_t$）估计。Giga–Sohr 只给 L_t^q $(1 < q < \infty)$；端点 $q = \infty$ 不成立，因为 $\int_0^t \|Ae^{\nu\tau A}\|\, d\tau = \infty$。修正：引理改为 L^q 形式，第四步取 $q = p$，得 $\nabla u \in L_t^p L^\infty \subset L_t^1$，正好满足 BKM。改后证明更短，也不再需要 δ-截断。

四、审计确认了什么（这些没有改）

Prodi–Serrin 与 BKM 的证明、Kato 不等式的方向与用法、二维 bootstrap 的整体思路、指定力定理的压力符号与 $\operatorname{div} f = 0$、代价定律的链式法则与 λ^3 标度、推论 7.3 的 $s = 1/2$ 阈值——都经两路独立复核通过。特别地，SymPy 的六项检查残差全为 0，干净目录重跑 `verify.py` 的输出与论文附录 B 逐位一致。

五、无法核验的部分

(1) CIV 与 IKX 两篇 2026 年定理本身：只核对了摘要与假设，没有重证。(2) 2026 年那份 166 页手稿与 Lean 工程：没有逐页审阅。(3) 数值只核验恒等式，不核验任何先验估计；T3 的差分误差 2.6×10^{-6} 决定了它看不见相对量级 10^{-6} 的错误。(4) 没有任何人类审稿人读过修订版。

六、给"AI 写的数学" 的审计协议

R1. 两个角色不同的审稿人：一个整体核对，一个拿着" 最可能出错清单" 逐点核对；不一致比" 一致" 更有信息量。

R2. 逐个引理问：右边出现的是数据的哪个范数？（F5、F6 都是这样暴露的）

R3. 不信任端点指数：凡是把 L_t^q 定理写成 $\sup_t$ 的地方都要重推（F7）。

R4. 所有指数用符号计算重推一遍（F12 的 3/2 因子，散文审稿永远看不出来）。

R5. 把" 数值验证过" 与" 已证明" 分开记账（F11）。

R6. 数值脚本换到空目录重跑。

R7. 每条被引用的定理都回到原始出处核对（F3）。

A Reproduction

All commands below were run inside the sandbox that produced this report.

```
# 1. clean-room numerics (paper/verify.py -> empty directory)
mkdir -p /tmp/cleanroom && cp paper/verify.py /tmp/cleanroom/
cd /tmp/cleanroom && python3 -u verify.py

# 2. symbolic checks (this report)
cd report && python3 -u symbolic_checks.py

# 3. rebuild paper, notes and this report
cd paper && tectonic -X compile ns-forced-vs-unforced.tex
```

```
cd ../notes && tectonic -X compile notes.tex
cd ../report && tectonic -X compile report.tex
```

Key numbers of the clean-room run (identical to Appendix B of the paper):

```
T1  max |div v| = 2.096e-13   max |residual| = 0   max |div f| = 1.511e-10
T2  lambda=2: 4.491e-12       lambda=3: 5.684e-12   (sup |R| = 1.029e+02)
T3  sup|div U| = 8.882e-16    sup|F| = 3.262263
    (T-t)^(3/2)||f|| = 1.153384 at t = 0.25, 0.5, 0.875
T4  |int (u.grad w) w| = 6.574e-15  (relative 5.017e-19)
```

B Symbolic verification output

The script `report/symbolic_checks.py` (SymPy) re-derives six identities from their definitions. Verbatim output:

```
======================================================================
C1  chain rule  d/dt[lam*U(lam*x)] = lamdot*(U + y U'),  y = lam*x
   U = exp(-Y**2/2)  ->  difference = 0
   U = Y**3 - Y  ->  difference = 0
   U = cos(Y)  ->  difference = 0
   lamdot / lam^3 = a (must be a)

======================================================================
C2  exponent bookkeeping  lam^(3-3/q) = (2a(T-t))^(-(3/2)(1-1/q))
   log-difference = 0 (must be 0)

======================================================================
C3  Kato-type inequality
    Delta|w|^p - p|w|^(p-2) w Delta w = p(p-1)|w|^(p-2)|grad w|^2
   p=2: residual = 0 (must be 0)
   p=4: residual = 0 (must be 0)

======================================================================
C4  Prodi-Serrin / Young exponents
   1/q + (q-2)/(2q) + 1/2      = 1
   (1-3/q)*(2q/(q-3))          = 2
   1/sigma + 1/sigma' , sigma=2q/(q+3), sigma'=2q/(q-3)  = 1

======================================================================
C5  vorticity of the Gaussian profile used in Cor. 7.3 / Thm 8.1
   curl u = (-x**2 - y**2 + 2)*exp(-x**2/2 - y**2/2)
   check  : 0

======================================================================
C6  2D vorticity identity  (u.grad)w = curl((u.grad)u)  for div-free u
   div u = 0
   (u.grad)w - curl((u.grad)u) = 0
```

References

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